Adult Census Income Classification

a. Problem statement

The objective is to predict whether an individual's annual income is above $50,000 using census attributes. This is a binary classification task.

b. Dataset description

The UCI Adult dataset contains 48,842 records and 14 predictor features. The official adult.data file (32,561 rows) is used for training and cross-validation. The official adult.test file (16,281 rows) is reserved for final evaluation. Missing numerical values are median-imputed; missing categorical values are mode-imputed. Categorical features are one-hot encoded and numerical features are standardized.

c. Github Repository Link

Add your GitHub repository URL here after uploading the project.

d. Models used and evaluation

The five required classifiers are tuned using stratified 3-fold cross-validation on the training partition. The selected models are evaluated once on the untouched official test partition. Hyperparameter selection uses F1.

Model	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8057	0.9032	0.5591	0.8398	0.6713	0.5637
Decision Tree	0.8078	0.8879	0.5616	0.8495	0.6762	0.5712
KNN	0.8437	0.8889	0.6935	0.6061	0.6469	0.5491
Naive Bayes	0.5880	0.8090	0.3568	0.9272	0.5153	0.3579
Support Vector Machine	0.7914	0.8948	0.5370	0.8495	0.6580	0.5469

Model performance observations

Logistic Regression: Accuracy=0.8057, AUC=0.9032, Precision=0.5591, Recall=0.8398, F1=0.6713, MCC=0.5637. A linear classification baseline. Its precision, recall and AUC show how effectively a linear decision boundary separates income classes.

Decision Tree: Accuracy=0.8078, AUC=0.8879, Precision=0.5616, Recall=0.8495, F1=0.6762, MCC=0.5712. Captures non-linear relationships and feature interactions. Depth and leaf-size tuning helps control overfitting.

KNN: Accuracy=0.8437, AUC=0.8889, Precision=0.6935, Recall=0.6061, F1=0.6469, MCC=0.5491. Uses distances to neighboring observations. Its results can be affected by feature scaling and the high-dimensional encoded feature space.

Naive Bayes: Accuracy=0.5880, AUC=0.8090, Precision=0.3568, Recall=0.9272, F1=0.5153, MCC=0.3579. Provides a fast probabilistic baseline. Its conditional-independence assumption may be restrictive when census attributes are related.

Support Vector Machine: Accuracy=0.7914, AUC=0.8948, Precision=0.5370, Recall=0.8495, F1=0.6580, MCC=0.5469. Uses an efficient linear maximum-margin classifier. Regularization tuning controls the trade-off between fitting the training data and generalization.

Overall winner: Decision Tree is the winner under the selected F1-first ranking. The final choice should also consider AUC, MCC, accuracy and the relative cost of false positives and false negatives.

Visual comparison of evaluation metrics

